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UNIVERSITY OF THE PUNJAB

BS (After Associate Degree) : Fifth Semester – Fall 2023

(Common with BS4Y 5th)

Paper: Classical Mechanics

Course Code: PHYS-3101 / PHY-301

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions: **(6x5=30)**

1. Show that the kinetic energy of a two-particle system is
$$\frac{1}{2}M\,\dot{R}^2 + \frac{1}{2}\mu v^2$$
where $M = m_1 + m_2$, v is the relative speed, and $\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}$(reduced mass).

2. A sphere of radius ρ is constrained to roll without slipping on the lower half of the inner surface of a hollow cylinder of inside radius R. Determine the Lagrange function, the equation of constraint and Lagrange's equation of motion. Find the frequency of small oscillation.

3. Explain virtual work. State D'Alembert's Principle and use it to derive the Lagrange's equation of motion
$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i} = Q_i\,.$$

4. Show that the transformation
$$q = \sqrt{2P}\sin Q,$$
$$p = \sqrt{2P}\cos Q,$$
is canonical.

5. Find x and y as functions of time, such that the functional
$$I = \int_{t_0}^{t_1}\left[\frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - mgy\right]dt$$
has stationary value.

6. Show that for a single particle with constant mass, the equation of motion implies
$$\frac{dT}{dt} = \mathbf{F}.\mathbf{v},$$
where T is the kinetic energy, $\mathbf{F}$ is the applied force vector and $\mathbf{v}$ is the velocity vector. If the mass varies with time, the corresponding equation is
$$\frac{d}{dt}(mT) = \mathbf{F}.\mathbf{p}.$$

Q.2. Answer the following questions. **(3x10=30)**

1. A particle of mass m moves in a central force field with potential $-\frac{k}{r}$. The Lagrangian is
$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\varphi}^2) + \frac{k}{r}$$
Find the momenta $(p_r, p_\theta, p_\varphi)$ conjugate to coordinates (r, θ, φ).Also find the Hamiltonian $H = H(r, \theta, \varphi, p_r, p_\theta, p_\varphi)$ and write down Hamilton's equations.

2. Find the force law for a central force field that allows a particle to move in a logarithmic spiral orbit given by $r = k\exp(\alpha\theta)$ where k and α are constants.

3. Show that the Poisson bracket obeys the Jacobi identity
$$\{A,\{B,C\}\} + \{B,\{C,A\}\} + \{C,\{A,B\}\} = 0$$
where A, B and C are arbitrary dynamical variables.

$$\Rightarrow F(r) = \mu(\ddot{r} - r\dot{\theta}^2) \quad \& \quad r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0$$

ADP

4.2: Two body central force problem & its reduction to one body problem:-

(Two body problem & reduced mass):- Consider a conserved system for two particles m_1 & m_2 . We shall limit ourselves to the case where the only forces acting are equal and opposite, directed along the line connecting the masses.

Suppose $\vec{r'}_1$ be position vector of m_1 w. r. t. center of mass and $\vec{r'}_2$ be position vector of m_2 w. r. t. center of mass, then

$$\vec{r} = \vec{r}_1 - \vec{r}_2 = \vec{r}_1' - \vec{r}_2'$$

$$\& \quad \vec{r}_1 = \vec{R} + \vec{r'}_1 \quad \& \quad \vec{r'}_2 = \vec{R} + \vec{r}_2'$$

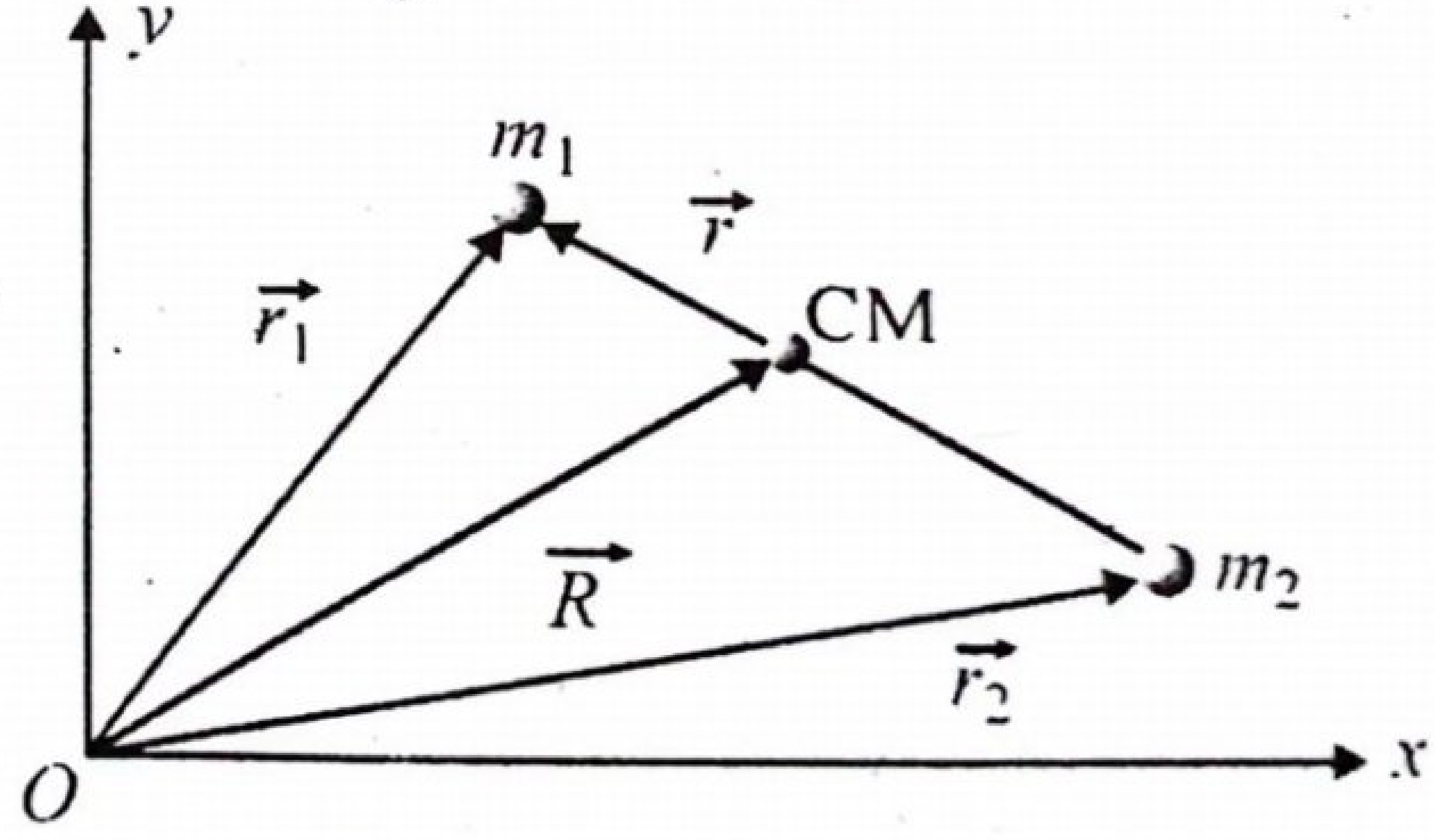

Fig 4:2: Two body problem

Position of center of mass is defined by,

$$\vec{R} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2}$$

By definition of center of mass,

$$m_1\vec{r}_1' + m_2\vec{r}_2' = 0 \quad \Rightarrow m_1\vec{r}_1' = -m_2\vec{r}_2'$$

$$\Rightarrow \vec{r}_1' = \frac{-m_2}{m_1}\vec{r}_2' \qquad \Rightarrow \vec{r}_2' = -\frac{m_1}{m_2}\vec{r}_1'$$

So $\vec{r} = \vec{r}_1' - \vec{r}_2'$ gives

$$\vec{r} = \vec{r}_1' - \left(-\frac{m_1}{m_2}\vec{r}_1'\right) \Rightarrow \vec{r} = \vec{r}_1' + \frac{m_1}{m_2}\vec{r}_1' \Rightarrow \vec{r} = \vec{r}_1'\left(1 + \frac{m_1}{m_2}\right)$$

$$\Rightarrow \vec{r} = \vec{r}_1'\left(\frac{m_1 + m_2}{m_2}\right) \qquad \Rightarrow \vec{r}_1' = \left(\frac{m_2}{m_1 + m_2}\right)\vec{r}$$

Now $\vec{r}_2' = -\frac{m_1}{m_2}\vec{r}_1' \Rightarrow \vec{r}_2' = -\frac{m_1}{m_2}\left(\frac{m_2}{m_1+m_2}\right)\vec{r} \quad \Rightarrow \vec{r}_2' = -\left(\frac{m_1}{m_1+m_2}\right)\vec{r}$

From $\vec{R} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1+m_2}$ we have,

$$\Rightarrow (m_1+m_2)\vec{R} = m_1\vec{r}_1 + m_2\vec{r}_2 \quad \Rightarrow m_1\vec{r}_1 = (m_1+m_2)\vec{R} - m_2\vec{r}_2$$

$$\Rightarrow m_1\vec{r}_1 = (m_1+m_2)\vec{R} - m_2(\vec{r}_1 - \vec{r}) \quad \Rightarrow m_1\vec{r}_1 = (m_1+m_2)\vec{R} - m_2\vec{r}_1 + m_2\vec{r}$$

$$\Rightarrow m_1\vec{r}_1 + m_2\vec{r}_1 = (m_1+m_2)\vec{R} + m_2\vec{r} \quad \Rightarrow \vec{r}_1(m_1+m_2) = (m_1+m_2)\vec{R} + m_2\vec{r}$$

$$\Rightarrow \vec{r}_1 = \vec{R} + \left(\frac{m_2}{m_1+m_2}\right)\vec{r}$$

Similarly,

$$\vec{r}_2 = \vec{R} - \left(\frac{m_1}{m_1+m_2}\right)\vec{r}$$

The Lagrangian L of system can written as,

$$L = \frac{1}{2}m_1\dot{r}_1^2 + \frac{1}{2}m_2\dot{r}_2^2 - V(r)$$

$$\Rightarrow L = \frac{1}{2}m_1\left\{\dot{\vec{R}} + \left(\frac{m_2}{m_1+m_2}\right)\dot{\vec{r}}\right\}^2 + \frac{1}{2}m_2\left\{\dot{\vec{R}} - \left(\frac{m_1}{m_1+m_2}\right)\dot{\vec{r}}\right\}^2 - V(r)$$

$$\Rightarrow L = \frac{1}{2}m_1\left\{\dot{R}^2 + \left(\frac{m_2}{m_1+m_2}\right)^2\dot{r}^2 + 2\left(\frac{m_2}{m_1+m_2}\right)\dot{\vec{R}}.\dot{\vec{r}}\right\}$$

$$+\frac{1}{2}m_2\left\{\dot{R}^2 + \left(\frac{m_1}{m_1+m_2}\right)^2\dot{r}^2 - 2\left(\frac{m_1}{m_1+m_2}\right)\dot{\vec{R}}.\dot{\vec{r}}\right\} - V(r)$$

$$\Rightarrow L = \frac{1}{2}m_1\dot{R}^2 + \frac{1}{2}m_1\dot{r}^2\left(\frac{m_2}{m_1+m_2}\right)^2 + m_1\left(\frac{m_2}{m_1+m_2}\right)\dot{\vec{R}}.\dot{\vec{r}} + \frac{1}{2}m_2\dot{R}^2$$

$$+\frac{1}{2}m_2\dot{r}^2\left(\frac{m_1}{m_1+m_2}\right)^2 - m_2\left(\frac{m_1}{m_1+m_2}\right)\dot{\vec{R}}.\dot{\vec{r}} - V(r)$$

$$\Rightarrow L = \frac{1}{2}m_1\dot{R}^2(m_1+m_2) + \frac{1}{2}\dot{r}^2 m_1m_2\left\{\left(\frac{m_2}{m_1+m_2}\right)^2 + \left(\frac{m_1}{m_1+m_2}\right)^2\right\}$$

$$+m_2\left(\frac{m_1}{m_1+m_2}\right)\dot{\vec{R}}.\dot{\vec{r}} - m_2\left(\frac{m_1}{m_1+m_2}\right)\dot{\vec{R}}.\dot{\vec{r}} - V(r)$$

$$\Rightarrow L = \frac{1}{2}(m_1+m_2)\dot{R}^2 + \frac{1}{2}\dot{r}^2\frac{m_1m_2}{(m_1+m_2)^2}(m_1+m_2) - V(r)$$

$$\Rightarrow L = \frac{1}{2}(m_1+m_2)\dot{R}^2 + \frac{1}{2}\dot{r}^2\frac{1}{\frac{(m_1+m_2)}{m_1m_2}} - V(r)$$

$$\Rightarrow L = \frac{1}{2}(m_1+m_2)\dot{R}^2 + \frac{1}{2}\dot{r}^2\frac{1}{\left(\frac{1}{m_1}+\frac{1}{m_2}\right)} - V(r)$$

$$\Rightarrow L = \frac{1}{2}M\dot{R}^2 + \frac{1}{2}\mu\dot{r}^2 - V(r)$$

where $M = m_1 + m_2$ & $\mu = \frac{1}{\frac{1}{m_1}+\frac{1}{m_2}}$

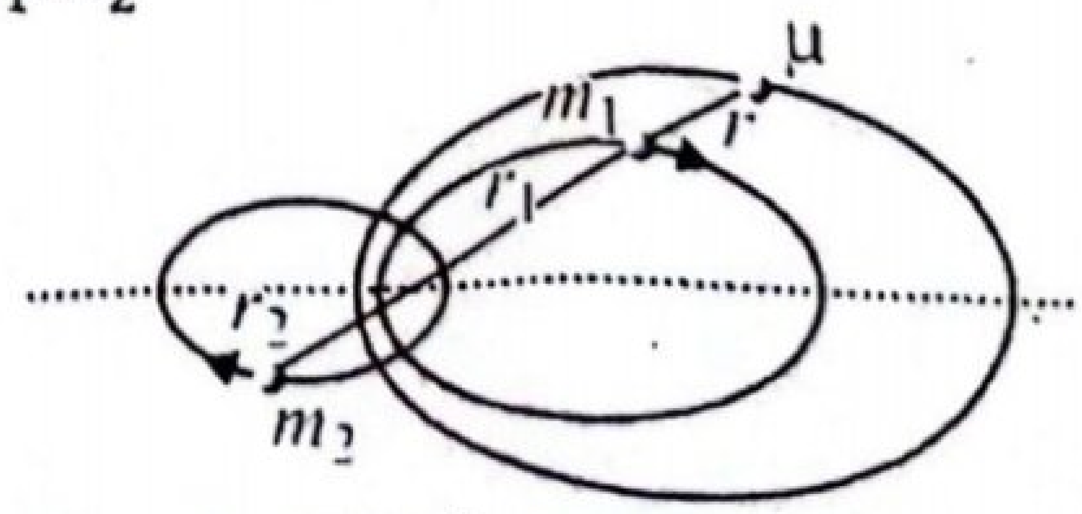

Fig 4.3: Effective potential and reduced mass

is called reduce mass. It is Lagrangian for a particle of mass μ moving in a central force field described by potential function V.

sphere and represent forces of constraints.

Example 3.28:- A sphere of radius r is constrained to roll without slipping on the lower half of the inner surface of a hollow cylinder of inside radius R. Determine the Lagrangian function, the equation of constraint, and Lagrange's equations of motion. Find the frequency of small oscillations.

Solution:-A sphere of radius r is constrained to roll without slipping on the lower half of the inner surface of a hollow cylinder of inside radius R is shown in fig 3.28. Generalized coordinates are θ *and* φ.

Kinetic energy is,

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$$T = \frac{1}{2}m\{(R-r)\dot{\theta}\}^2 + \frac{1}{2}I\dot{\varphi}^2$$

where I is the moment of inertia of sphere with respect to its diameter and $I = \frac{2}{5}mr^2$, so

$$T = \frac{1}{2}m\{(R-r)\dot{\theta}\}^2 + \frac{1}{2}\frac{2}{5}mr^2\dot{\varphi}^2$$

$$\Rightarrow T = \frac{1}{2}m\{(R-r)\dot{\theta}\}^2 + \frac{1}{5}mr^2\dot{\varphi}^2$$

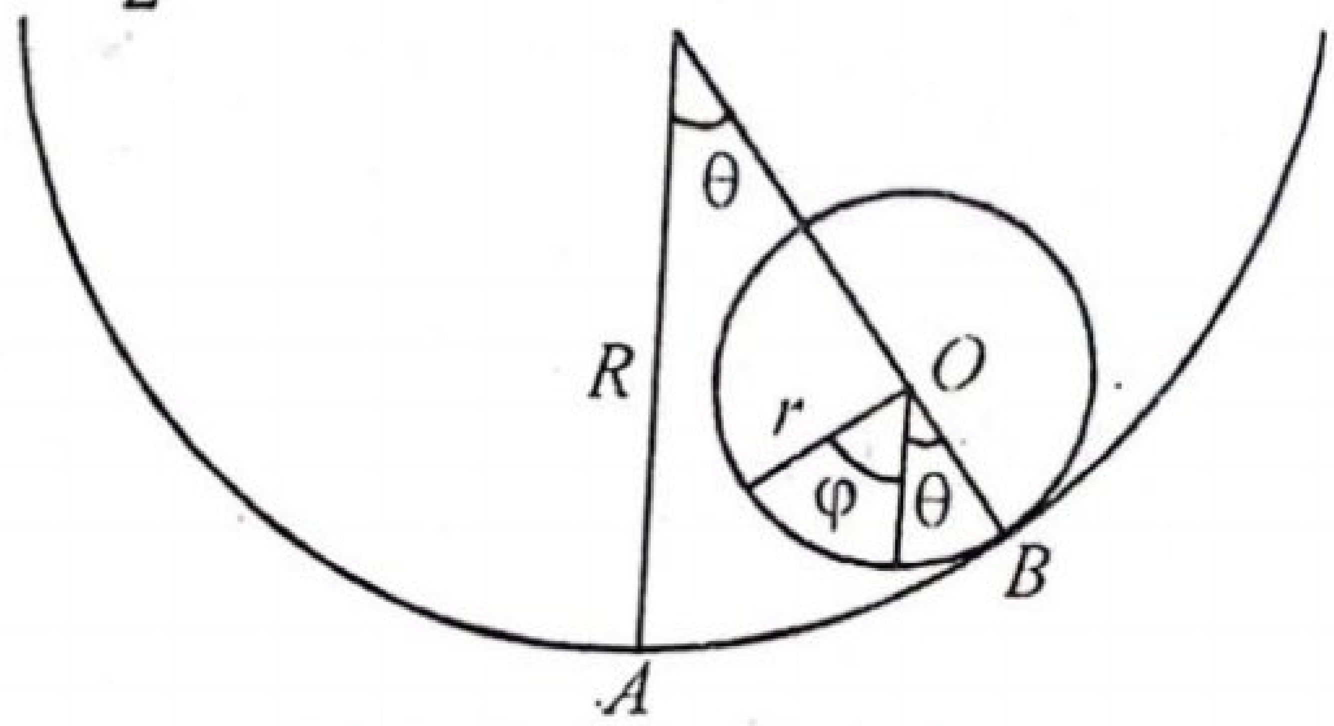

Fig 3.33: Example 3.28

Potential energy is,

$$V = \{R - (R-r)\cos\theta\}mg$$

Lagrangian is,

$$L = \frac{1}{2}m(R-r)^2\dot{\theta}^2 + \frac{1}{5}mr^2\dot{\varphi}^2 - \{R - (R-r)\ \cos\theta\}mg \quad \rightarrow (a)$$

When the sphere is at its lowest position, the points A and B coincide. The condition $AO = BO$ gives the equation of constraint:

$$(R-r)\theta - r\varphi = 0 \quad \rightarrow (b)$$

Therefore, we have two Lagrange's equations with one undetermined multiplier:

$$\left.\begin{aligned}\frac{\partial L}{\partial \theta} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) + \lambda\frac{\partial f}{\partial \theta} = 0\\ \frac{\partial L}{\partial \varphi} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\varphi}}\right) + \lambda\frac{\partial f}{\partial \varphi} = 0\end{aligned}\right\} \quad \rightarrow (c)$$

From (a), $\frac{\partial f}{\partial \theta} = R - r$ and $\frac{\partial f}{\partial \varphi} = -r$ so equations (c) give,

$$-(R-r)mg\sin\theta - m(R-r)^2\ddot{\theta} + \lambda(R-r) = 0 \quad \rightarrow (d)$$

$$-\frac{2}{5}mr^2\ddot{\varphi} - \lambda r = 0 \quad \rightarrow (e)$$

From (e),

$$\lambda = -\frac{2}{5}mr\ddot{\varphi} \quad \rightarrow (f)$$

Using equation (b), we have

$$\lambda = -\frac{2}{5}m(R-r)\ddot{\theta} \quad \rightarrow (g)$$

Putting (g) into (d), we obtain,

$$-(R-r)mg\sin\theta - m(R-r)^2\ddot{\theta} + \frac{2}{5}m(R-r)\ddot{\theta} \times (R-r) = 0$$

$$\Rightarrow \ddot{\theta} + \left(\frac{5g}{7(R-r)}\right)\sin\theta = 0 \quad \Rightarrow \ddot{\theta} + \omega^2\sin\theta = 0$$

where ω is the frequency of small oscillations, defined by,

$$\omega = \sqrt{\frac{5g}{7(R-r)}}$$

Practice Problem 3.28:- *Consider a block of mass m sliding on an inclined plane*

3:-They are to be carried out in a manner consistent with constraints.

2.4.2: Principle of virtual work:- ADP

According to this principle,

"If for any arbitrary virtual displacement $\delta\vec{r}_i$, the virtual work of forces of constraints vanishes, the virtual work of external applied forces $\vec{F}_i^e$ also vanishes and the system is in equilibrium".

Proof:-Consider a sclernomic system of N particles. If the system is in equilibrium, then total force on each particle must be zero $\vec{F}_i = 0$ for each i and evidently $\vec{F}_i.\,\delta\vec{r}_i$, which is virtual work done on ith particle by $\vec{F}_i$ in a virtual displacement $\delta\vec{r}_i$ is also zero. The total virtual work on system is,

$$\sum_i \vec{F}_i.\,\delta\,\vec{r}_i = 0 \qquad \Rightarrow \sum_i (\vec{F}_i^e + \vec{f}_i).\,\delta\,\vec{r}_i = 0$$

$$\Rightarrow \sum_i \vec{F}_i^e.\,\delta\,\vec{r}_i + \sum_i \vec{f}_i.\,\delta\,\vec{r}_i = 0$$

Many of the constraints that commonly occurs such as sliding motion on a frictionless surface and rolling contact without slipping, do no work under a virtual displacement that is;

$$\vec{f}_i\,.\,\delta\,\vec{r}_i = 0$$

This is practically only possible when we consider forces of constraints $\vec{f}_i$ perpendicular to $\delta\vec{r}_i$. If we exclude friction forces from consideration and restrict over selves to a system for which the virtual work of forces of constraints vanishes that is;

$$\sum_i \vec{f}_i.\,\delta\,\vec{r}_i = 0$$

Then

$$\sum_i \vec{F}_i^e.\,\delta\,\vec{r}_i = 0$$

which is principle of virtual work.

Example 2.10:- A uniform plank of mass M is

$$\Rightarrow h = \frac{}{\sqrt{M^2 - m^2}} \quad \text{as required.}$$

ADP

2.5: D' Alembert Principle:-The principle of virtual work deals with *statics*. *We want* to find a principle that involves the general motion of system. Such a principle was suggested by Bernoulli and then developed by D' Alembert.

From Newton second law of motion,

$$\vec{F}_i = \frac{d\vec{p}_i}{dt} \Rightarrow \vec{F}_i = \dot{\vec{p}}_i \quad \Rightarrow \vec{F}_i - \dot{\vec{p}}_i = 0$$

If we regard '$-\dot{\vec{p}}_i$' as a force, an inertial or reversed effective force as named by Bernoulli and D' Alembert, which added to $\vec{F}_i$ produces equilibrium, then dynamics reduces to statics and we have;

$$\sum_i (\vec{F}_i - \dot{\vec{p}}_i).\delta\vec{r}_i = 0$$

As $\vec{F}_i = \vec{F}_i^e + \vec{f}_i$, so

$$\sum_i (\vec{F}_i^e + \vec{f}_i - \dot{\vec{p}}_i).\delta\vec{r}_i = 0 \quad \Rightarrow \sum_i (\vec{F}_i^e - \dot{\vec{p}}_i).\delta\vec{r}_i + \sum_i \vec{f}_i\,.\delta\vec{r}_i = 0 = 0$$

If we restrict ourselves to a system for which virtual work of forces of constraints vanishes, then

$$\sum_i (\vec{F}_i^e - \dot{\vec{p}}_i).\delta\vec{r}_i = 0$$

This equation is Lagrange form of D' Alembert principle. The superscript 'e' is dropped and $\vec{F}_i$ will mean applied force on ith particle that is;

$$\sum_i (\vec{F}_i - \dot{\vec{p}}_i).\delta\vec{r}_i = 0$$

$$W_1 \sin\theta\,\delta\theta - W_2 \sin\phi\,\delta\theta = 0 \quad \Rightarrow W_1 \sin\theta = W_2 \sin\phi \quad \Rightarrow \frac{\;}{W_2} = \frac{\;}{\sin\theta}$$

ADP

2.5.1 Derivation of Lagrange equation from D' Alembert principle:- D' Alembert principle is,

$$\sum_i (\vec{F}_i - \dot{\vec{p}}_i).\delta\vec{r}_i = 0 \quad \Rightarrow \sum_i \vec{F}_i \cdot \delta\vec{r}_i - \sum_i \dot{\vec{p}}_i.\delta\vec{r}_i = 0$$

$$\Rightarrow \sum_i \vec{F}_i \cdot \delta\vec{r}_i = \sum_i \dot{\vec{p}}_i.\delta\vec{r}_i \quad \rightarrow (2.2)$$

Considering right side,

$$\sum_i \dot{\vec{p}}_i.\delta\vec{r}_i = \sum_i \frac{d}{dt}(m_i\vec{v}_i).\delta\vec{r}_i = \sum_i \frac{d}{dt}\left(m_i\frac{d\vec{r}_i}{dt}\right).\delta\vec{r}_i = \sum_i \frac{d}{dt}(m_i\dot{\vec{r}}_i).\delta\vec{r}_i = \sum_i m_i\ddot{\vec{r}}_i.\delta\vec{r}_i$$

Put $\delta\vec{r}_i = \sum_j \frac{\partial\vec{r}_i}{\partial q_j}\delta q_j$ we get

$$\sum_i \dot{\vec{p}}_i.\delta\vec{r}_i = \sum_i m_i\ddot{\vec{r}}_i \cdot \sum_j \frac{\partial\vec{r}_i}{\partial q_j}\delta q_j \quad \Rightarrow \sum_i \dot{\vec{p}}_i.\delta\vec{r}_i = \sum_i\sum_j m_i\ddot{\vec{r}}_i \cdot \frac{\partial\vec{r}_i}{\partial q_j}\delta q_j \quad \rightarrow (2.3)$$

Now,

$$\frac{d}{dt}\left(\dot{\vec{r}}_i\cdot\frac{\partial\vec{r}_i}{\partial q_j}\right) = \ddot{\vec{r}}_i\cdot\frac{\partial\vec{r}_i}{\partial q_j} + \dot{\vec{r}}_i\cdot\frac{d}{dt}\left(\frac{\partial\vec{r}_i}{\partial q_j}\right) \Rightarrow \frac{d}{dt}\left(\dot{\vec{r}}_i\cdot\frac{\partial\vec{r}_i}{\partial q_j}\right) - \dot{\vec{r}}_i\cdot\frac{\partial}{\partial q_j}\left(\frac{d\vec{r}_i}{dt}\right) = \ddot{\vec{r}}_i\cdot\frac{\partial\vec{r}_i}{\partial q_j}$$

So equation (2.3) becomes;

$$\sum_i \dot{\vec{p}}_i\cdot\delta\vec{r}_i = \sum_i\sum_j m_i\left\{\frac{d}{dt}\left(\dot{\vec{r}}_i\cdot\frac{\partial\vec{r}_i}{\partial q_j}\right) - \dot{\vec{r}}_i\cdot\frac{\partial}{\partial q_j}\left(\frac{d\vec{r}_i}{dt}\right)\right\}\delta q_j$$

$$\Rightarrow \sum_i \dot{\vec{p}}_i\cdot\delta\vec{r}_i = \sum_i\sum_j m_i\left\{\frac{d}{dt}\left(\dot{\vec{r}}_i\cdot\frac{\partial\vec{r}_i}{\partial q_j}\right) - \dot{\vec{r}}_i\cdot\frac{\partial\dot{\vec{r}}_i}{\partial q_j}\right\}\delta q_j \quad \rightarrow (2.4)$$

From relation

$$\vec{r}_i = \vec{r}_i(q_1, q_2, \dots\dots\dots, q_n, t)$$

$$\Rightarrow d\vec{r}_i = \frac{\partial\vec{r}_i}{\partial q_1}dq_1 + \frac{\partial\vec{r}_i}{\partial q_2}dq_2 + \cdots + \frac{\partial\vec{r}_i}{\partial q_n}dq_n + \frac{\partial\vec{r}_i}{\partial t}dt \Rightarrow d\vec{r}_i$$

$$= \sum_j \frac{\partial\vec{r}_i}{\partial q_j}dq_j + \frac{\partial\vec{r}_i}{\partial t}dt$$

$$\Rightarrow \frac{d\vec{r}_i}{dt} = \sum_j \frac{\partial\vec{r}_i}{\partial q_j}\frac{dq_j}{dt} + \frac{\partial\vec{r}_i}{\partial t} \quad \Rightarrow \dot{\vec{r}}_i = \sum_j \frac{\partial\vec{r}_i}{\partial q_j}\dot{q}_j + \frac{\partial\vec{r}_i}{\partial t} \quad \Rightarrow \boxed{\frac{\partial\dot{\vec{r}}_i}{\partial\dot{q}_j} = \frac{\partial\vec{r}_i}{\partial q_j}}$$

So equation (2.4) becomes,

$$\sum_i \dot{\vec{p}}_i\cdot\delta\vec{r}_i = \sum_i\sum_j m_i\left\{\frac{d}{dt}\left(\dot{\vec{r}}_i\cdot\frac{\partial\dot{\vec{r}}_i}{\partial\dot{q}_j}\right) - \dot{\vec{r}}_i\cdot\frac{\partial\dot{\vec{r}}_i}{\partial q_j}\right\}\delta q_j$$

$$\Rightarrow \sum_i \dot{\vec{p}}_i\cdot\delta\vec{r}_i = \sum_i\sum_j m_i\left\{\frac{d}{dt}\frac{\partial}{\partial\dot{q}_j}\left(\frac{\dot{r}_i^2}{2}\right) - \frac{\partial}{\partial q_j}\left(\frac{\dot{r}_i^2}{2}\right)\right\}\delta q_j$$

$$\Rightarrow \sum_i \dot{\vec{p}}_i\cdot\delta\vec{r}_i = \sum_j\left\{\frac{d}{dt}\frac{\partial}{\partial\dot{q}_j}\left(\sum_i\frac{1}{2}m_i\dot{r}_i^2\right) - \frac{\partial}{\partial q_j}\left(\sum_i\frac{1}{2}m_i\dot{r}_i^2\right)\right\}\delta q_j$$

$$\Rightarrow \sum_i \dot{\vec{p}}_i\cdot\delta\vec{r}_i = \sum_j\left\{\frac{d}{dt}\frac{\partial}{\partial\dot{q}_j}\left(\sum_i T_i\right) - \frac{\partial}{\partial q_j}\left(\sum_i T_i\right)\right\}\delta q_j = \sum_j\left\{\frac{d}{dt}\left(\frac{\partial T}{\partial\dot{q}_j}\right) - \frac{\partial T}{\partial q_j}\right\}\delta q_j$$

Using equation (2.2), we have

$$\sum_i \vec{F}_i\cdot\delta\vec{r}_i = \sum_j\left\{\frac{d}{dt}\left(\frac{\partial T}{\partial\dot{q}_j}\right) - \frac{\partial T}{\partial q_j}\right\}\delta q_j$$

$$\Rightarrow \sum_j Q_j\delta q_j = \sum_j\left\{\frac{d}{dt}\left(\frac{\partial T}{\partial\dot{q}_j}\right) - \frac{\partial T}{\partial q_j}\right\}\delta q_j$$

$$\Rightarrow \frac{d}{dt}\left(\frac{\partial T}{\partial\dot{q}_j}\right) - \frac{\partial T}{\partial q_j} = Q_j \quad \rightarrow (2.5)$$

where $j = 1, 2, 3 \dots n$, being degrees of freedom. Equations (2.5) are Lagrange equation.

ADP **Example 6.12:-** Show that the transformation $q = \sqrt{2P}\sin Q$, $p = \sqrt{2P}\cos Q$ is canonical by Poisson's brackets.

Solution:-Squaring and adding q and p, we get

$$q^2 + p^2 = 2Psin^2Q + 2Pcos^2Q \quad \Rightarrow q^2 + p^2 = 2P \quad \Rightarrow P = \frac{1}{2}(q^2 + p^2)$$

Dividing q by p, we get

$$\frac{q}{p} = \frac{\sqrt{2P}\sin Q}{\sqrt{2P}\cos Q} = \tan Q \quad \Rightarrow Q = \tan^{-1}\frac{q}{p}$$

We know that Poisson bracket of two dynamical variables F and G is defined,

$$[F, G] = \frac{\partial F}{\partial q}\frac{\partial G}{\partial p} - \frac{\partial F}{\partial p}\frac{\partial G}{\partial q}$$

Now,

$$[P, P] = \frac{\partial P}{\partial q}\frac{\partial P}{\partial p} - \frac{\partial P}{\partial p}\frac{\partial P}{\partial q} = 0$$

Similarly,

$$[Q, Q] = 0$$

Poisson bracket of two functions F & G is defined as,

$$[Q, P] = \frac{\partial Q}{\partial q}\frac{\partial P}{\partial p} - \frac{\partial Q}{\partial p}\frac{\partial P}{\partial q}$$

$$\frac{\partial Q}{\partial q} = \frac{p}{p^2 + q^2}, \quad \frac{\partial Q}{\partial p} = \frac{-q}{p^2 + q^2}, \quad \frac{\partial P}{\partial p} = p \quad , \quad \frac{\partial P}{\partial q} = q$$

So $$[Q, P] = \frac{p \times p}{p^2 + q^2} + \frac{q \times q}{p^2 + q^2} \quad \Rightarrow [Q, P] = \frac{p^2 + q^2}{p^2 + q^2} \quad \Rightarrow [Q, P] = 1$$

Hence the transformation $q = \sqrt{2P}\sin Q$, $p = \sqrt{2P}\cos Q$ is canonical by Poisson's brackets.

Practice Problem 3.14:- *Find x and y as functions of time when the functional I has stationary value.*

$$I = \int_{t_0}^{t} \left\{\frac{1}{2}m\left(\dot{x}^2 + \dot{y}^2\right) - mgy\right\} dt$$

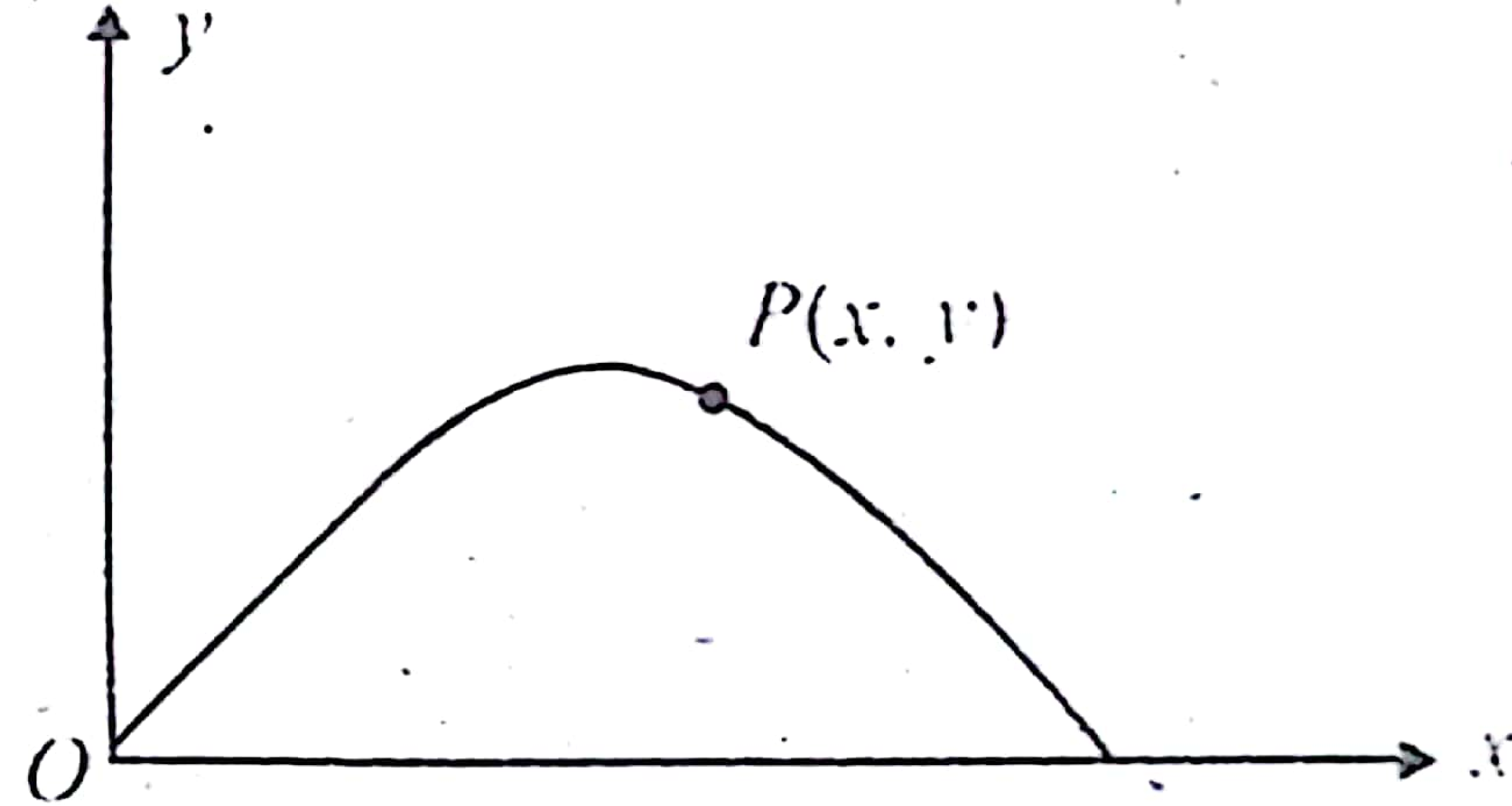

Fig 3.14: Practice Problem 3.14

Hint:-Let,

$$L = f = \frac{1}{2}m\left(\dot{x}^2 + \dot{y}^2\right) - mgy$$

Lagrange equations are,

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = 0 \quad \text{and} \quad \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{y}}\right) - \frac{\partial L}{\partial y} = 0$$

$$\Rightarrow \frac{d}{dt}(m\dot{x}) - 0 = 0 \quad \text{and} \quad \frac{d}{dt}(m\dot{y}) + mg = 0$$

$$\Rightarrow m\ddot{x} = 0 \quad \text{and} \quad m\ddot{y} + mg = 0 \quad \Rightarrow \ddot{x} = 0 \quad \text{and} \quad \ddot{y} = -g$$

$$\Rightarrow \dot{x} = c_1 \quad \text{and} \quad \dot{y} = -gt + d_1 \quad \Rightarrow x = c_1 t + c_2 \quad \text{and} \quad y = -\frac{gt^2}{2} + d_1 t + d_2$$

Example 3.15:- (a)-A particle of mass m moves in one dimension such ...

motion of individual particle relative to centre of mass.

Example 1.19:- **(a)-Show that for a single particle with constant mass, the equation of motion implies**

$$\frac{dT}{dt} = \vec{F}\,.\,\vec{v}$$

where T is kinetic energy, $\vec{F}$ is applied force vector and $\vec{v}$ is the velocity vector.
(b) If the mass varies with time, the corresponding equation is,

$$\frac{d}{dt}(mT) = \vec{F}\cdot\vec{p}$$

Solution:- (a)-Kinetic energy is,

$$T = \frac{1}{2}mv^2 = \frac{1}{2}m\vec{v}\cdot\vec{v}$$

Now,

$$\frac{dT}{dt} = \frac{1}{2}m\left(\vec{v}\cdot\frac{d\vec{v}}{dt} + \frac{d\vec{v}}{dt}\cdot\vec{v}\right) \quad \Rightarrow \frac{dT}{dt} = \frac{1}{2}m\left(\vec{v}\cdot\frac{d\vec{v}}{dt}\vec{v}\cdot\frac{d\vec{v}}{dt}\right) \quad \Rightarrow \frac{dT}{dt} = \frac{1}{2}m\left(2\vec{v}\cdot\frac{d\vec{v}}{dt}\right)$$

$$\Rightarrow \frac{dT}{dt} = m\vec{v}.\vec{a} \qquad \Rightarrow \frac{dT}{dt} = m\vec{a}.\vec{v} \qquad \Rightarrow \frac{dT}{dt} = \vec{F}.\vec{v}$$

(b)-Kinetic energy is,

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$$T = \frac{1}{2}mv^2 = \frac{m^2v^2}{2m} \quad \Rightarrow T = \frac{p^2}{2m} \quad \Longrightarrow mT = \frac{p^2}{2} \quad \Rightarrow mT = \frac{1}{2}(\vec{p}.\vec{p})$$

Differentiating w. r. t. 't',

$$\frac{d}{dt}(mT) = \frac{1}{2}\left(\vec{p}\cdot\frac{d\vec{p}}{dt} + \frac{d\vec{p}}{dt}\cdot\vec{p}\right) \quad \Rightarrow \frac{d}{dt}(mT) = \frac{1}{2}\left(\vec{p}\cdot\frac{d\vec{p}}{dt} + \vec{p}\cdot\frac{d\vec{p}}{dt}\right)$$

$$\Rightarrow \frac{d}{dt}(mT) = \frac{2}{2}\left(\vec{p}\cdot\frac{d\vec{p}}{dt}\right)$$

$$\frac{d}{dt}(mT) = \vec{p}.\frac{d\vec{p}}{dt} = \vec{p}.\vec{F} \quad \Rightarrow \frac{d}{dt}(mT) = \vec{F}.\vec{p}$$ as required.

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L.Q #1

Example 5.21:- **A particle moves in a spherically symmetric force field with potential energy given by,**

$$V(r) = -\frac{k}{r}$$

Calculate the Hamiltonian function in spherical coordinates, and obtain the canonical equations of motion.

Solution:-To solve this problem we make use of spherical polar coordinates. Kinetic energy is,

$$T = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2 sin^2\dot{\theta}\varphi^2\right)$$

Potential energy is,

$$V(r) = -\frac{k}{r}$$

Lagrangian is,

$$L = T - V = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2 sin^2\dot{\theta}\varphi^2\right) + \frac{k}{r}$$

Conjugate momenta to generalized coordinates r, θ, and φ are,

$$p_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r} \;\Rightarrow\; \dot{r} = \frac{p_r}{m} \quad \& \quad p_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} \;\Rightarrow\; \dot{\theta} = \frac{p_\theta}{mr^2}$$

$$p_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = mr^2\dot{\varphi}sin^2\theta \;\Rightarrow\; \dot{\varphi} = \frac{p_\varphi}{mr^2 sin^2\theta}$$

Hamiltonian is,

$$H = T + V = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\dot{\varphi}^2 sin^2\theta\right) - \frac{k}{r}$$

$$\Rightarrow H = \frac{1}{2}\left(\frac{{p_r}^2}{m} + \frac{{p_\theta}^2}{mr^2} + \frac{{p_\varphi}^2}{mr^2 sin^2\theta}\right) - \frac{k}{r}$$

The canonical equations of motion are,

$$\dot{p}_r = -\frac{\partial H}{\partial r} = -\frac{k}{r^2} + \frac{{p_\theta}^2}{mr^3} + \frac{{p_\varphi}^2}{mr^3 sin^2\theta}, \quad \dot{p}_\theta = -\frac{\partial H}{\partial \theta} = \frac{{p_\varphi}^2 \cot\theta}{mr^2 sin^2\theta} \quad \& \quad \dot{p}_\varphi = -\frac{\partial H}{\partial \varphi} = 0$$

The last equation shows that p_φ = constant.

Practice Problem 5.21: Consider a partic...

$$= \dot{\theta}\,\frac{dr}{d\theta}$$

$$\dot{\theta} = \frac{l}{\mu r^2}$$

$$\frac{dr}{dt} = \frac{l}{\mu r^2}\frac{dr}{d\theta}$$

$$\frac{l}{\mu r^2}\frac{dr}{d\theta} = \sqrt{\frac{2}{\mu}\left(E - V - \frac{l^2}{2\mu r^2}\right)}$$

$$d\theta = \frac{l\,dr}{\mu r^2\sqrt{\frac{2}{\mu}\left(E - V - \frac{l^2}{2\mu r^2}\right)}}$$

$$d\theta = \frac{l\,dr}{r^2\sqrt{2\mu\left(E - V - \frac{l^2}{2\mu r^2}\right)}}$$

L.Q #2

Law of force is given by

$$r = K e^{\alpha\theta}$$

where $K > 0$, $\alpha > 0$ are const

(a) Find Law Of force.

Law of force is given by

$$f(r) = -\frac{l^2}{\mu r^2}\left[\frac{d^2}{d\theta^2}u + u\right]$$

$$r = k e^{\alpha\theta}$$

$$u = \frac{1}{k} e^{-\alpha\theta}$$

$$\frac{du}{d\theta} = -\frac{\alpha}{k} e^{-\alpha\theta}$$

$$\frac{d^2u}{d\theta^2} = \frac{\alpha^2}{k} e^{-\alpha\theta}$$

$$f(r) = \frac{l^2}{\mu k^2 e^{2\alpha\theta}} \left[e^{-\alpha\theta} \cdot \frac{\alpha^2}{k} + \frac{e^{-\alpha\theta}}{k} \right]$$

$$= \frac{l^2}{\mu k^3 e^{3\alpha\theta}} (\alpha^2 + 1)$$

$$= \frac{l^2}{\mu r^3} (\alpha^2 + 1)$$

$$f(r) = \frac{l^2 (\alpha^2 + 1)}{\mu} \cdot \frac{1}{r^3}$$

(b) Find r and θ as a fn of time

$$l = \mu r^2 \dot{\theta}$$

$$\dot{\theta} = \frac{l}{\mu r^2}$$

$$\dot{\theta} = \frac{l}{\mu k^2} e^{-2\alpha\theta}$$

$$\frac{d\theta}{dt} = \frac{l}{\mu k^2} e^{-2\alpha\theta}$$

$$\int e^{2\alpha\theta} \, d\theta = \int \frac{l}{\mu k^2} \, dt$$

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L.Q #3

Example 6.23:- Prove the Poisson bracket obeys the Jacobi identity;

$$[A,[B,C]]+[B,[C,A]]+[C,[A,B]]=0$$

Proof:-Consider,

$$[A,[B,C]]=\sum_i\left\{\frac{\partial A}{\partial q_i}\frac{\partial[B,C]}{\partial p_i}-\frac{\partial A}{\partial p_i}\frac{\partial[B,C]}{\partial q_i}\right\}$$

$$\Rightarrow[A,[B,C]]=\sum_i\left\{\frac{\partial A}{\partial q_i}\frac{\partial}{\partial p_i}\left(\frac{\partial B}{\partial q_i}\frac{\partial C}{\partial p_i}-\frac{\partial B}{\partial p_i}\frac{\partial C}{\partial q_i}\right)-\frac{\partial A}{\partial p_i}\frac{\partial}{\partial q_i}\left(\frac{\partial B}{\partial q_i}\frac{\partial C}{\partial p_i}-\frac{\partial B}{\partial p_i}\frac{\partial C}{\partial q_i}\right)\right\}$$

$$\Rightarrow[A,[B,C]]=\sum_i\left\{\frac{\partial A}{\partial q_i}\left(\frac{\partial^2 B}{\partial p_i\partial q_i}\frac{\partial C}{\partial p_i}+\frac{\partial B}{\partial q_i}\frac{\partial^2 C}{\partial p_i^2}-\frac{\partial^2 B}{\partial p_i^2}\frac{\partial C}{\partial q_i}-\frac{\partial B}{\partial p_i}\frac{\partial^2 C}{\partial p_i\partial q_i}\right)\right.$$

$$\left.-\frac{\partial A}{\partial p_i}\left(\frac{\partial^2 B}{\partial q_i^2}\frac{\partial C}{\partial p_i}+\frac{\partial B}{\partial q_i}\frac{\partial^2 C}{\partial q_i\partial p_i}-\frac{\partial^2 B}{\partial p_i\partial q_i}\frac{\partial C}{\partial q_i}-\frac{\partial B}{\partial p_i}\frac{\partial^2 C}{\partial q_i^2}\right)\right\}\rightarrow(a)$$

Similarly

$$[B,[C,A]] = \sum_i \left\{ \frac{\partial B}{\partial q_i}\frac{\partial^2 C}{\partial p_i \partial q_i}\frac{\partial A}{\partial p_i} + \frac{\partial B}{\partial q_i}\frac{\partial C}{\partial q_i}\frac{\partial^2 A}{\partial p_i^2} - \frac{\partial B}{\partial q_i}\frac{\partial^2 C}{\partial p_i^2}\frac{\partial A}{\partial q_i} - \frac{\partial B}{\partial q_i}\frac{\partial C}{\partial p_i}\frac{\partial^2 A}{\partial p_i \partial q_i} \right.$$
$$\left. - \frac{\partial B}{\partial p_i}\frac{\partial^2 C}{\partial q_i^2}\frac{\partial A}{\partial p_i} - \frac{\partial B}{\partial p_i}\frac{\partial C}{\partial q_i}\frac{\partial^2 A}{\partial q_i \partial p_i} + \frac{\partial B}{\partial p_i}\frac{\partial^2 C}{\partial p_i \partial q_i}\frac{\partial A}{\partial q_i} + \frac{\partial B}{\partial p_i}\frac{\partial C}{\partial p_i}\frac{\partial^2 A}{\partial q_i^2} \right\} \rightarrow (b)$$

$$[C,[A,B]] = \sum_i \left\{ \frac{\partial C}{\partial q_i}\frac{\partial^2 A}{\partial p_i \partial q_i}\frac{\partial B}{\partial p_i} + \frac{\partial C}{\partial q_i}\frac{\partial A}{\partial q_i}\frac{\partial^2 B}{\partial p_i^2} - \frac{\partial C}{\partial q_i}\frac{\partial^2 A}{\partial p_i^2}\frac{\partial B}{\partial q_i} - \frac{\partial C}{\partial q_i}\frac{\partial A}{\partial p_i}\frac{\partial^2 B}{\partial p_i \partial q_i} \right.$$
$$\left. - \frac{\partial C}{\partial p_i}\frac{\partial^2 A}{\partial q_i^2}\frac{\partial B}{\partial p_i} - \frac{\partial C}{\partial p_i}\frac{\partial A}{\partial q_i}\frac{\partial^2 B}{\partial q_i \partial p_i} + \frac{\partial C}{\partial p_i}\frac{\partial^2 A}{\partial p_i \partial q_i}\frac{\partial B}{\partial q_i} + \frac{\partial C}{\partial p_i}\frac{\partial A}{\partial p_i}\frac{\partial^2 B}{\partial q_i^2} \right\} \rightarrow (c)$$

Adding equations (a), (b) and (c), we obtain

$$[A,[B,C]] + [B,[C,A]] + [C,[A,B]] = 0$$

According to this theorem